

1 Acoustic Emission and Passive Acoustic Regression to
2 the ISO 281
3 Viscosity Ratio κ for Online, Non-Invasive Lubrication
4 Condition Monitoring of Ball Bearings: An Exploratory
5 Study

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10 **Keywords:** acoustic emission; lubrication condition monitoring; viscosity ratio κ ;
11 ball bearing; signal processing; gradient boosting; sensor fusion

Abstract

Inadequate lubrication accounts for an estimated 30–80% of rolling element bearing failures, yet reliable online, non-invasive lubrication condition monitoring (LCM) remains unsolved. This paper presents an exploratory study in which acoustic emission (AE) and passive ultrasound (UL) signals are simultaneously acquired from a ball bearing during a continuous speed sweep (1–5985 rpm) at temperatures spanning 33–100 °C, covering boundary, mixed, and full-film lubrication regimes across 8 783 sweep–sensor pairs per channel. The ISO 281 viscosity ratio $\kappa \in [0.003, 1.64]$ serves as a continuous, physics-grounded regression target. LightGBM on AE features alone achieves the best hold-out performance ($R^2 = 0.951$, RMSE = 0.078 κ -units). Sensor fusion (AE+UL) provides only a marginal improvement that is not confirmed by hold-out significance testing. The dataset is made publicly available to support standardised comparison.

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1 Introduction

Inadequate lubrication is estimated to account for between 30% and 80% of all rolling element bearing failures [1–3]. Despite this, lubrication condition monitoring (LCM) has received substantially less research attention than vibration-based fault detection. Accurate, online LCM would enable condition-based relubrication rather than fixed-schedule approaches that produce both under- and over-lubrication [4].

Passive, non-invasive sensing methods—acoustic emission (AE), vibration, and passive ultrasound (US)—are attractive for industrial LCM because they require no bearing modification and can be retrofitted. AE and passive ultrasound at 38–40 kHz are both in practical use, but have not been studied simultaneously in a unified regression framework with a physics-based lubrication reference variable.

The ISO 281 viscosity ratio κ has been used as a continuous regression target [1, 5]: a model trained to predict κ implicitly learns the operating-condition dependence and produces a physically interpretable output. Whether jointly exploiting AE and passive US improves κ prediction over either alone is an open question with direct practical implications.

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Contributions

1. **Novel dual-channel experiment.** The first study to simultaneously acquire AE and passive US signals from a ball bearing and regress both, individually and jointly, to κ using a common dataset and analytical ground truth.
2. **Sub-band feature extraction for AE.** Features are computed from three frequency sub-bands (20–500 kHz, 500 kHz–1 MHz, 1–2 MHz), enabling model-driven identification of the most informative AE frequency range.
3. **Quantified fusion test.** The first quantitative test of AE–US complementarity in a unified experimental framework.
4. **Rigorous statistical evaluation.** Model comparisons use repeated nested cross-validation and corrected significance tests that properly account for data re-use.

The remainder of the paper is organised as follows. Section 2 reviews the theoretical background. Section 3 describes the experimental setup. Section 4 presents the signal processing and feature extraction pipeline. Section 5 presents the feature analysis results. Section 6 presents the regression modelling and statistical evaluation. Section 7 discusses the findings. Section 8 concludes.

2 Background

2.1 Lubrication regimes and the viscosity ratio κ

The ISO 281 standard defines the viscosity ratio

$$\kappa = \frac{\nu_0}{\nu_1(n, d_{pw})}, \quad (1)$$

where ν_0 [mm²s⁻¹] is the actual kinematic viscosity at operating temperature and ν_1 is the minimum required for full EHL at speed n and pitch diameter d_{pw} [6]:

$$\nu_1 = \begin{cases} 45\,000 \, n^{-0.83} \, d_{pw}^{-0.5} & n < 1000 \text{ rpm}, \\ 4\,500 \, n^{-0.5} \, d_{pw}^{-0.5} & n \geq 1000 \text{ rpm}. \end{cases} \quad (2)$$

A value of $\kappa \geq 1$ indicates adequate film formation; $\kappa < 0.5$ corresponds to severe boundary lubrication. Using κ as a regression target means a trained model implicitly compensates for operating-condition effects.

2.2 Physical origins of AE and passive ultrasound signals

AE refers to transient elastic stress waves generated by asperity contact and micro-fracture in rolling bearings; emission rate, amplitude, and frequency content are all sensitive to the lubrication regime [7, 8]. AE sensors typically operate up to 1 MHz and require sampling rates exceeding 500 kHz.

Passive ultrasound sensing at 38–40 kHz provides a complementary, lower-frequency view. Commercial instruments compare the broadband ultrasonic level against a baseline, triggering relubrication when the level rises by a specified number of decibels [9]. In the present experiment the UL channel uses a heterodyned probe, limiting usable bandwidth to approximately 10 kHz. Whether the partial independence of the two modalities translates into complementary κ information is the central empirical question of this study.

3 Experimental Setup

In this section the test rig for conducting experiments of bearing runs with varying lubrication conditions and the sensing setup are described. A single experiment was conducted at CeramicSpeed in Herning, Denmark, during which the operating conditions were varied in terms of temperature and rotational speed, such that a wide range of κ values were covered.

3.1 Test rig and bearing

The test rig was a custom-built bearing test stand at CeramicSpeed (Holstebro, Denmark) built for running a bearing under different operating conditions. A single deep-groove ball bearing (model SYJ25, bore 25 mm, pitch diameter $d_{pw} = 38.0$ mm, 9 chrome steel rolling elements, dynamic load rating 14000 N) was driven by a VFD-controlled electric motor. The bearing was lubricated by a defined quantity of Keratech 22 oil ($\nu_{40} = 22$ cSt, $\nu_{100} = 4.1$ cSt). Temperature was monitored by a PT100 sensor on the bearing housing.

Note on VFD noise: The VFD which drives the test rig is known to introduce significant electrical noise. This was sought mitigated by a filter.

3.2 Sensor configuration and data acquisition

Two sensors are acquired simultaneously:

AE sensor. A wideband piezoelectric AE sensor bonded to the bearing housing. Sampling rate: 1.6 MHz; effective bandwidth verified to 800 kHz.

Passive UL sensor. A heterodyned passive ultrasound probe clamped to the bearing housing. The heterodyning process limits effective content to below 10 kHz. Sampling rate: 1.6 MHz.

Both channels share identical acquisition timing, ensuring co-temporal signal segments.

3.3 Operating conditions and κ values

The experiment follows a continuous speed-sweep protocol: the VFD ramps the shaft from rest to approximately 6000 rpm, with bearing temperature rising naturally from 33 °C to 100 °C. After signal quality filtering, 8783 sweeps per sensor channel are retained. Table ?? summarises the operating envelope and resulting κ distribution.

The κ value for each sweep is computed via Equations (1)–(2) using measured RPM and temperature, with ν_0 evaluated via the ASTM D341 Walther viscosity–temperature equation. No separate film-thickness reference measurement is available; κ is therefore model-derived subject to the assumptions of the Hamrock–Dowson framework.

4 Signal Processing Pipeline

4.1 Signal cleaning

Each sweep–sensor pair is subject to the following quality checks before feature extraction: short/zero/NaN sweeps are rejected; isolated non-finite samples are interpolated; clipping and saturation are flagged; outlier samples exceeding six standard deviations are replaced by the local median. After cleaning, 1598 of 1600 sweep–sensor pairs are retained.

What is the amount?

Do we need NLGI consistency grade?

Write more about the filter and its effect on the signals.

How and where is it mounted?

4.2 Waveform and spectral visualisation

4.3 Feature extraction

Twenty-six features are computed from each cleaned sweep-sensor signal:

Time-domain (12): peak, RMS, standard deviation, variance, skewness, kurtosis, crest factor, shape factor, impulse factor, margin factor, Hjorth mobility, Hjorth complexity.

Frequency-domain (14): dominant frequency, spectral mean, spectral standard deviation, spectral skewness, spectral kurtosis, spectral flatness (Wiener entropy), centre frequency, RMS frequency, frequency-weighted standard deviation, peak frequency, normalised frequency standard deviation, frequency skewness, frequency kurtosis, and normalised bandwidth.

AE sub-band decomposition. The full 26-feature set is extracted independently for each of three zero-phase Butterworth bandpass sub-bands: 20–500 kHz, 500 kHz–1 MHz, and 1–2 MHz, yielding $26 \times 4 = 104$ candidate AE features.

UL channel pre-filtering. The UL channel is pre-filtered to 0–20 kHz before broadband feature extraction, giving 26 candidate UL features.

5 Feature Analysis

5.1 Correlation with κ

Table 1 lists the top ten features per sensor ranked by $|\rho(\text{feature}, \kappa)|$. The top AE features are drawn overwhelmingly from the 1–2 MHz and 500 kHz–1 MHz sub-bands (peak $|\rho| = 0.891$), while UL amplitude features reach at most $|\rho| = 0.663$. A two-stage selection retains 32 AE and 6 UL features after a joint Spearman–Pearson threshold filter and a greedy VIF and pairwise-correlation redundancy reduction.

Add example waveforms and spectra here to illustrate the raw signals and the effect of cleaning.

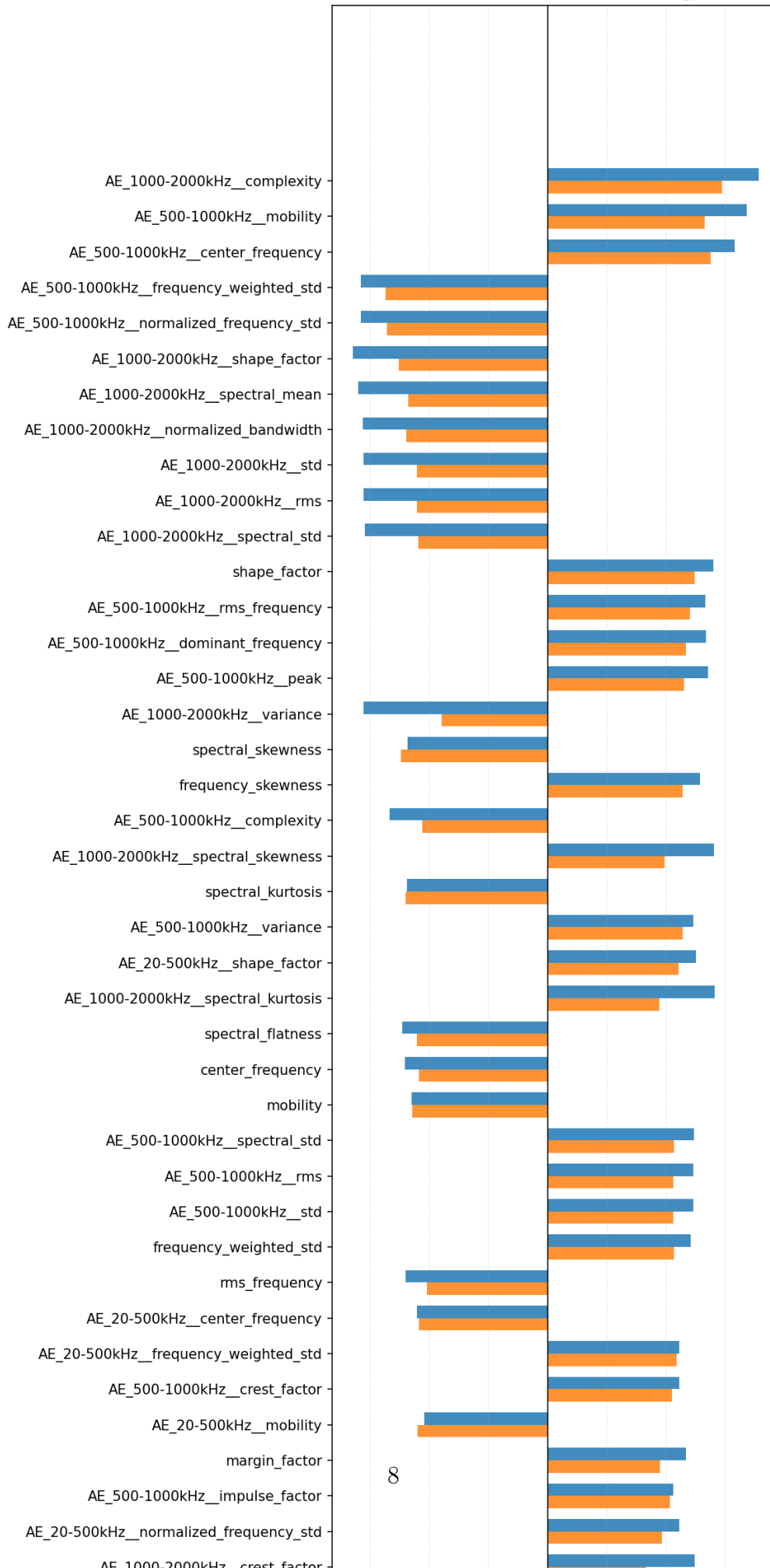
Show the sub-bands and explain the information in those.

Make table with features and refer to papers where they

Table 1: Top 10 features by combined Spearman–Pearson rank score with κ , for the AE channel (all sub-bands and broadband combined) and the UL channel. After the Stage 2 redundancy reduction, 32 AE and 6 UL features are retained for modelling; the retained sets are dominated by sub-band features for AE and amplitude/spectral features for UL.

AE (best feature per rank)			UL	
Feature	$ \rho $	$ r $	Feature	$ \rho $
1–2 MHz complexity	0.891	0.736	Spectral mean	0.663
500k–1 MHz mobility	0.841	0.663	Peak	0.629
500k–1 MHz centre freq.	0.789	0.687	RMS	0.607
500k–1 MHz freq.-wt. std	0.789	0.686	Std	0.606
500k–1 MHz norm. freq. std	0.789	0.679	Spectral std	0.607
1–2 MHz shape factor	0.825	0.628	Variance	0.606
1–2 MHz spectral mean	0.801	0.590	Norm. bandwidth	0.607
1–2 MHz norm. bandwidth	0.782	0.599	Freq. skewness	0.539
1–2 MHz std	0.777	0.552	Freq. kurtosis	0.537
1–2 MHz RMS	0.777	0.552	Kurtosis	0.469

AE — feature correlation ranking



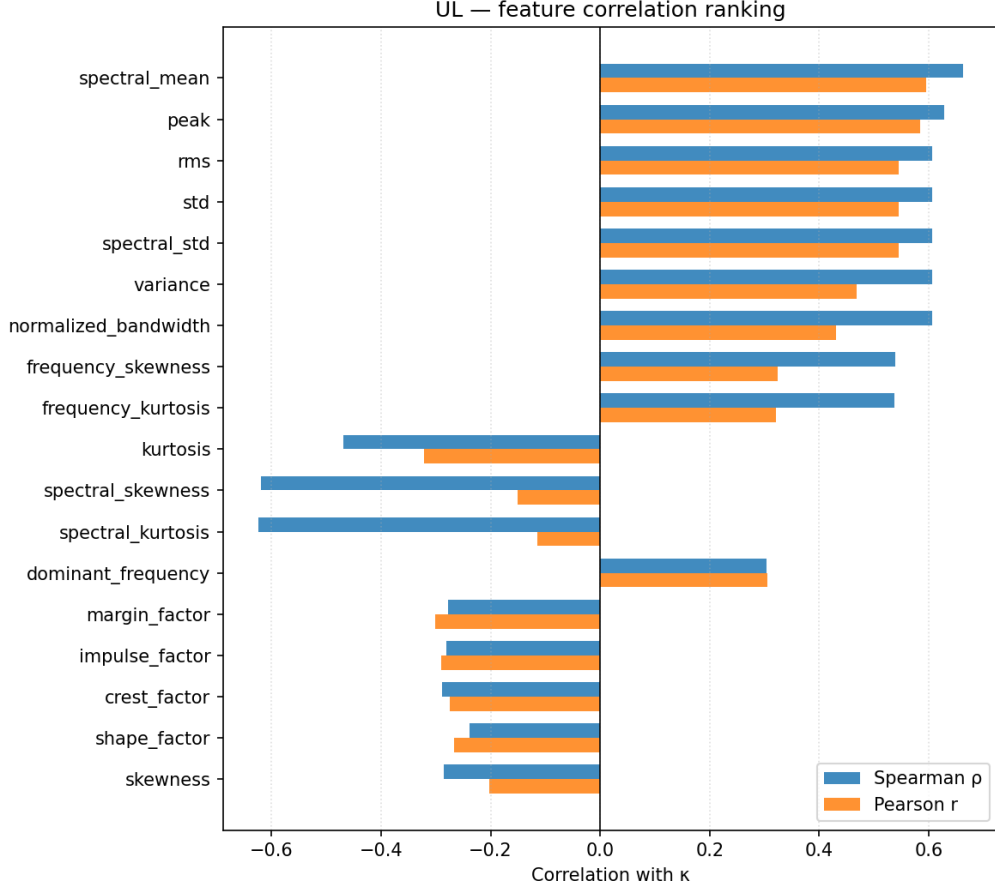


Figure 2: Spearman $|\rho|$ and Pearson $|r|$ between each UL-channel feature and κ , sorted by combined rank. Retained features highlighted.

5.2 Operating-condition confounding

RPM correlates with κ ($\rho = +0.62$) and temperature correlates with κ ($\rho = -0.68$), since both drive κ through ν_1 and ν_0 respectively while simultaneously affecting signal energy. Amplitude-dominated features (RMS, variance, peak) are therefore partly acting as indirect proxies for speed rather than direct indicators of lubrication film adequacy. The κ regression framework partially addresses this: training on the full multi-condition dataset allows the model to implicitly discriminate amplitude changes due to lubrication from those due to operating conditions.

5.3 PCA and regime separability

Figure 3 shows the first two principal components of the retained AE feature set, coloured by κ regime. Boundary and full-film sweeps are broadly separated in the PC1–PC2 plane, while mixed regime sweeps overlap with both, confirming that the mixed lubrication regime is the most challenging region.

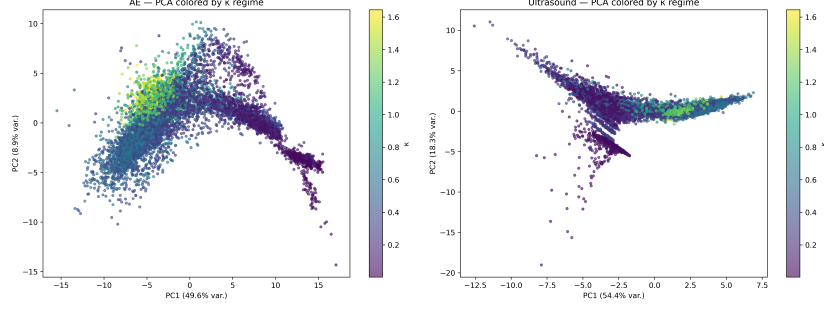


Figure 3: PCA projection of the retained AE feature set, coloured by lubrication regime. Boundary ($\kappa < 0.5$): red; mixed ($0.5 \leq \kappa < 1$): orange; full film ($\kappa \geq 1$): green.

6 Regression Modelling and Statistical Evaluation

6.1 Experimental design and holdout split

The dataset is divided into an 80% training set and a 20% hold-out test set by regime-stratified random sampling; the hold-out set is withheld from all model development steps. Three regression model families are evaluated for three sensor configurations (AE only, UL only, AE+UL):

1. **Elastic Net** with α and ℓ_1 ratio selected by `ElasticNetCV`.
2. **Degree-2 Polynomial Regression** (polynomial expansion + Ridge via `RidgeCV`).
3. **LightGBM** with hyperparameters tuned by Bayesian optimisation (Optuna TPE, 20 trials per inner fold).

6.2 Repeated nested cross-validation

Model evaluation uses repeated nested five-fold cross-validation ($R = 5$ repetitions), yielding $5 \times 5 = 25$ outer RMSE scores per model–sensor combination. Each outer fold contains its own inner optimisation loop for LightGBM, ensuring outer scores are not contaminated by hyperparameter selection on the outer fold.

6.3 Statistical significance testing

Pairwise significance tests are conducted at two levels.

Level 1 — Within-sensor comparisons (architecture differences). For pairs of models trained on the same sensor configuration (e.g., Elastic Net vs. Polynomial on AE), the corrected repeated k -fold t -test of Nadeau and Bengio [10] is applied to the 25 paired RMSE scores. This test corrects the standard paired t -test for the positive correlation

induced by shared training data across repeated folds. The correction factor is:

$$\hat{\sigma}_{\text{corr}}^2 = \left(\frac{1}{n_{\text{test}}} + \frac{n_{\text{train}}}{n_{\text{test}}} \right) \hat{\sigma}^2, \quad (3)$$

where $\hat{\sigma}^2$ is the sample variance of pairwise RMSE differences and $n_{\text{test}}/n_{\text{train}}$ is the test/train size ratio. Multiple within-sensor comparisons are adjusted by the Holm–Bonferroni step-down procedure at $\alpha = 0.05$.

Level 2 — Cross-sensor comparisons (best per sensor configuration). For the best model from each sensor configuration (selected by lowest mean CV RMSE), pairwise differences on the common hold-out set are assessed by:

- The Wilcoxon signed-rank test on paired residuals (non-parametric);
- The modified Diebold–Mariano test [11] for equal predictive accuracy under squared-error loss;
- A bootstrap Δ RMSE 95% confidence interval (10 000 resamples with replacement).

Cross-sensor p -values are also corrected by Holm–Bonferroni.

6.4 Results

Table 2: Regression model performance. CV RMSE: mean $\pm$ std of outer RMSE across 25 repeated nested CV folds ($n_{\text{train}} = 7\,026$ per fold). HO: hold-out test set ($n = 1\,757$). Best hold-out R^2 in bold. Bayesian Ridge (BR) was trained on the full training set and evaluated on hold-out only (not included in the nested CV framework) and is shown for reference.

Model	Sensors	CV RMSE	HO R^2	HO MAE	HO RMSE
LightGBM	Combined	0.0767 ± 0.0040	0.953	0.050	0.076
LightGBM	AE	0.0780 ± 0.0037	0.951	0.051	0.078
LightGBM	UL	0.2233 ± 0.0060	0.600	0.155	0.222
Elastic Net	Combined	0.1556 ± 0.0051	0.830	0.111	0.145
Elastic Net	AE	0.1568 ± 0.0052	0.826	0.113	0.147
Polynomial	Combined	0.1568 ± 0.0048	0.803	0.113	0.156
Polynomial	AE	0.1646 ± 0.0044	0.775	0.123	0.167
Polynomial	UL	0.2577 ± 0.0056	0.477	0.193	0.254
Elastic Net	UL	0.2783 ± 0.0068	0.396	0.210	0.273
BR (ref.)	Combined	—	0.847	0.103	0.137
BR (ref.)	AE	—	0.830	0.110	0.145
BR (ref.)	UL	—	0.536	0.181	0.239

Overall performance. LightGBM achieves the best hold-out performance across all sensor configurations (AE-only: $R^2 = 0.951$, RMSE = 0.078 κ -units, $\approx 15\%$ of target

range). Linear and polynomial models on AE features achieve $R^2 \approx 0.77\text{--}0.83$, confirming a non-linear feature- κ mapping.

Sensor comparison. AE features substantially outperform UL-channel features across all model families (LightGBM AE-UL gap: $\Delta R^2 = 0.35$, Nadeau-Bengio $p < 10^{-22}$, $\Delta \text{RMSE} = 0.144$, 95% CI $[0.133, 0.157]$).

Sensor fusion yields a marginal, statistically unconfirmed improvement. LightGBM on the combined AE+UL feature set achieves $R^2 = 0.953$ vs. AE-only 0.951. Hold-out tests do not confirm the improvement: Wilcoxon $p = 0.88$, Diebold-Mariano $p = 0.28$, bootstrap $\Delta \text{RMSE} = 0.0011$ with 95% CI $[-0.0009, 0.0033]$ spanning zero.

Model consistency. Close agreement between nested CV RMSE and hold-out RMSE across all models indicates stable generalisation with no evidence of significant overfitting.

Figure 4 shows predicted vs. actual κ for the hold-out set for all model families under the AE-only configuration.

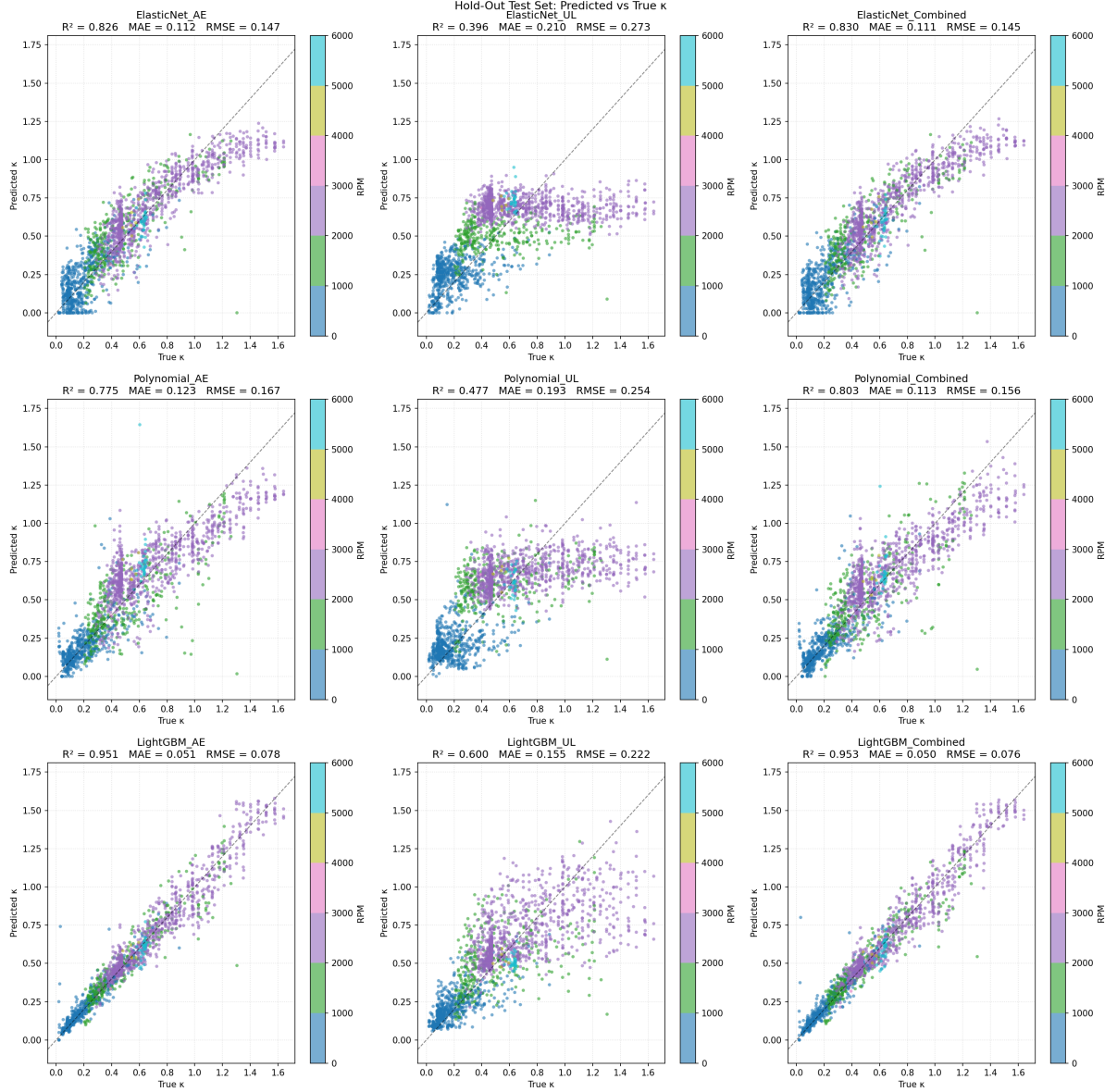


Figure 4: Predicted vs. actual κ on the hold-out test set for all model families and sensor configurations. The dashed diagonal indicates perfect prediction. Residual spread increases in the mixed lubrication regime ($0.5 \leq \kappa < 1.0$).

SHAP-based feature importance. The dominant contributor to LightGBM predictions is *1–2 MHz complexity* (mean $|\text{SHAP}| = 0.162$), followed by *500 kHz–1 MHz centre frequency* (0.051) and *1–2 MHz mobility* (0.038). The 1–2 MHz complexity feature is also the only feature ranked in the top 10 across all three model types (Table 3).

Table 3: Cross-model SHAP agreement for AE features: features appearing in the top 10 by mean $|\text{SHAP}|$ for more than one model type.

Feature	ElasticNet	Polynomial	LightGBM	Models in top 10
1–2 MHz complexity	1	1	1	3
1–2 MHz mobility	2	—	3	2
500k–1 MHz spectral std	3	—	10	2
Broadband freq. skewness	10	—	4	2
500k–1 MHz centre freq.	—	10	2	2
500k–1 MHz dominant freq.	—	4	7	2

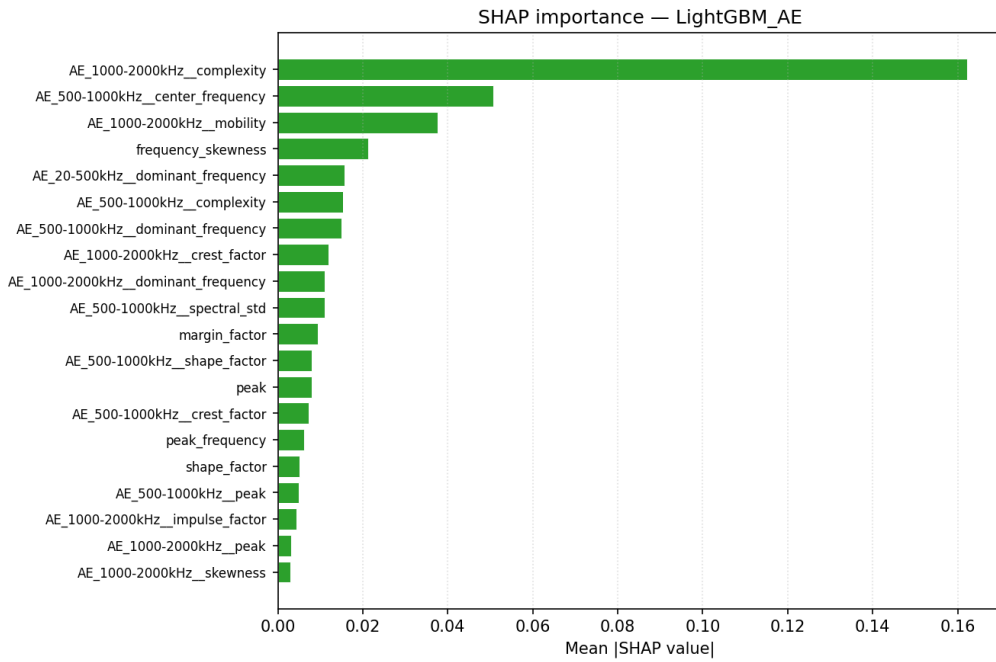


Figure 5: SHAP feature importance (mean $|\text{SHAP value}|$) for the LightGBM model trained on AE features. Features from frequency sub-bands are prefixed with the band label (e.g., AE_20–500kHz_).

7 Discussion

7.1 Why AE outperforms the UL channel

The AE sensor captures genuine wideband stress-wave emissions up to 800 kHz, directly sensitive to asperity contact events, while the heterodyned UL channel is limited to below 10 kHz where the signal is dominated by low-frequency machinery noise and VFD harmonics. Replacing the heterodyned probe with a wideband passive acoustic sensor would be expected to substantially improve UL-channel model performance.

7.2 Why fusion provides only marginal gain

The UL channel retains only 6 features after selection (vs. 32 for AE), with peak correlation $|\rho| \leq 0.66$ vs. $|\rho| = 0.89$ for AE, and its effective bandwidth partially overlaps with the low-frequency tail of the AE signal. Together these factors limit the independent κ information the UL channel can contribute. Whether a wideband passive ultrasound sensor would provide more substantial fusion gain remains an open question.

7.3 Interpreting the amplitude-confounding result

Amplitude features exploit speed-mediated amplitude changes to predict κ , which is operationally useful but may limit generalisation to operating conditions outside the training distribution. Future feature engineering should prioritise spectral shape and ratio features (frequency kurtosis, normalised bandwidth, spectral flatness) that are expected to carry a higher proportion of genuine lubrication-state information.

7.4 Implications for the κ regression framework

A hold-out $R^2 = 0.951$ and $\text{RMSE} = 0.078$ κ -units for a single AE sensor represents strong performance across a demanding speed-and-temperature ramp spanning boundary through full-film conditions, substantially exceeding prior comparable studies [1]. The repeated nested cross-validation design provides appropriately conservative estimates, and corrected significance tests confirm that performance differences are assessed under assumptions that properly account for data re-use.

7.5 Limitations

1. **Single bearing and lubricant.** Generalisability to other bearing types, sizes, or lubricants is unknown.
2. **Heterodyned UL channel.** Results should not be equated with wideband ultrasound.
3. **Regime imbalance.** Boundary lubrication dominates the dataset (60%); full-film conditions account for only 11%.
4. **VFD harmonic contamination.** Features below 10 kHz are affected by switching artefacts.
5. **Model-derived κ reference.** No independent film-thickness measurement is available.
6. **Single session file.** Temporal autocorrelation between adjacent sweeps may reduce effective independence between training and hold-out sets.

8 Conclusion

This paper has presented an exploratory study of simultaneous AE and passive ultrasound sensing regressed to κ for non-invasive lubrication condition monitoring. The main findings are:

1. **AE sub-band features are strong predictors of κ .** LightGBM on 32 retained AE features achieves hold-out $R^2 = 0.951$, RMSE = 0.078 κ -units; the 1–2 MHz complexity feature is the dominant and most consistent predictor across model types.
2. **Sensor fusion provides only a marginal, statistically unconfirmed improvement.** The LightGBM combined model achieves $\Delta R^2 = +0.002$ over AE alone, not confirmed by hold-out significance tests.
3. **AE strongly and significantly outperforms the UL channel** ($\Delta R^2 = 0.35$ for LightGBM), reflecting the fundamental bandwidth limitation of the heterodyned UL probe.
4. **The κ regression framework is validated** as a highly effective approach for passive non-invasive LCM.

Future work should address the identified limitations by acquiring wideband passive ultrasound, validating across multiple independent sessions and bearing types, and developing physically motivated sub-band ratio features robust to operating-condition variation.

CRedit authorship contribution statement

Frederik Appel Vardinghus-Nielsen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interests

The author declares that there are no competing interests.

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Data availability

The complete dataset (HDF5 signal files, extracted features in Parquet format, and the reproducible Python analysis pipeline) will be deposited on Zenodo upon acceptance [DOI to be assigned].

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted writing tools for language editing and structural drafting. After using these tools, the author reviewed and edited the content as necessary and takes full responsibility for the content of the publication.

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